

QUESTION BANK

Course Code & Name	23IT040 – Deep Learning
Year / Semester	III / V
Department	Information Technology
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Unit 1: Deep Networks Basics

S.No	Questions	CO	BL
Part A			
1.	Differentiate Artificial Intelligence and Machine Learning.	CO1	K2
2.	Define the term Vector and Scalar with an example.	CO1	K1
3.	What is a matrix and Tensor?	CO1	K1
4.	State the term Bernoulli's Distribution.	CO1	K1
5.	List the role of an Optimization.	CO1	K1
6.	State the advantages of Batch Gradient Descent.	CO1	K1
7.	Define Mini-Batch Gradient Descent.	CO1	K1
8.	What are the different instances of Gradient descent-based Optimizers?	CO1	K1
9.	List some specific areas where machine learning is being used.	CO1	K1
10.	Differentiate Over-fitting and under-fitting.	CO1	K2
11.	What is Hyperparameter and highlight its categories?	CO1	K1
12.	List the Role of Learning Rate.	CO1	K1
13.	State the difference between Model and Hyper parameters.	CO1	K2
14.	What are the methods used for Cross-Validation?	CO1	K1
15.	Define Estimator and its types.	CO1	K1
16.	How does Regularization Work in Machine Learning?	CO1	K1
17.	List the series of steps involved in Machine Learning lifecycle project.	CO1	K1
18.	Define the term Bias and Variance.	CO1	K1
19.	Highlight the Challenges in optimization techniques.	CO1	K1
20.	What is Feed-Forward Neural Network?	CO1	K1
Part B			
1.	Explain Linear Algebra in detail along with several types of mathematical objects.	CO1	K2
2.	Describe Probability Distributions and their classifications with appropriate graphical representations.	CO1	K2
3.	Illustrate gradient-based optimization techniques used in training deep networks.	CO1	K2
4.	A real estate company has collected a dataset containing 15,000 house records with features such as location, area (sq. ft.), number of bedrooms, number of bathrooms, age of the house, parking spaces, and selling price. The dataset is divided into 80% training data and 20% testing data to	CO1	K2

	develop a machine learning model for predicting house prices. Two different regression models are trained on the dataset. - Model A achieves 99% training accuracy with very low training error, but its performance on the testing dataset is poor, with high prediction error. - Model B achieves 65% training accuracy and also performs poorly on the testing dataset, showing high errors on both datasets. Based on the above dataset and model performance, answer the following: - Define overfitting and underfitting. - Identify which model represents overfitting and which represents underfitting with suitable justification. - Compare the characteristics of overfitting and underfitting in terms of: - Training accuracy - Testing accuracy - Explain the impact of overfitting and underfitting on machine learning model performance using the given housing price prediction dataset.		
5.	Using the following sample record, perform one iteration of SGD. House Area (1000 sq.ft) =2, Actual Price = 80 Lakhs Assume: - Initial weight $w = 30$ - Bias $b = 10$ - Learning rate $\alpha = 0.05$ Calculate: - Predicted house price - Prediction error - Gradient - Updated values of weight and bias	CO1	K2
6.	A social media company has collected a dataset containing 2 million user posts from platforms such as Twitter, Facebook, and Instagram. Each post is labelled into one of the following categories: - Sports - Politics - Entertainment - Technology - Health - Business The dataset consists of the following information:	CO1	K2

	Feature Description PostID Unique identifier Number of Likes Total likes received Number of shares Total shares Number of Comments Total Comments Hashtags Keywords associated with the post Category Target class The company aims to build an automatic text classification system capable of accurately classifying newly generated social media posts. The dataset is extremely large, contains noisy text, abbreviations, emojis, slang, and multilingual content, and is continuously updated with thousands of new posts every minute. Based on the above dataset, answer the following: 1. Explain the challenges faced by conventional machine learning algorithms in classifying large-scale social media text data. 2. Discuss how deep learning techniques overcome these challenges. 3. Compare conventional machine learning and deep learning with respect to: - o Feature extraction - o Scalability - o Handling unstructured text - o Accuracy - o Computational requirements - o Model adaptability		
7.	Discuss in detail about deep feed-forward neural network architecture and its work strategy. Highlight its applications.	CO1	K2
8.	An e-commerce company is building a product recommendation system. The team experiments with different hyperparameter values and uses a validation set to choose the best model. One model is too simple and another is too complex. The final model is selected based on its prediction performance. Explain the roles of hyperparameters, validation sets, estimators, bias, and variance in the development of the recommendation system.	CO1	K2

Unit 2: Convolutional Neural Networks

S.No	Questions	CO	BL
Part A			

1.	What is CNN and List the three types of layers in Neural Network.	CO2	K1
2.	How does a convolution layer extract features from an image?	CO2	K1
3.	List the layers used to build Convolutional Neural Network .	CO2	K1
4.	Define the term Convolution Operation.	CO2	K1
5.	What are Sparse Interactions and how feature map is calculated in it?	CO2	K1
6.	Define the term Stride and Padding.	CO2	K1
7.	What is shared parameter in Convolution Network?	CO2	K1
8.	What is pooling in a Convolutional Neural Network (CNN)? List the types of pooling layers with suitable feature map examples.	CO2	K1
9.	Define Convolution Variants and list its types.	CO2	K1
10.	What are the commonly used nonlinear activation functions in CNN?	CO2	K1
11.	List the Applications of CNN.	CO2	K1
12.	Differentiate between Cost Function vs Loss Function.	CO2	K2
13.	What is Regression Loss?	CO2	K1
14.	What is a transposed convolution? State one application.	CO2	K1
15.	What is Classification Loss and list its types.	CO2	K1
16.	What is Loss Function in Deep Learning?	CO2	K1
17.	What are the advantages of Dilated Convolution?	CO2	K1
18.	Differentiate between Max Pooling and Average Pooling.	CO2	K2
19.	Define Fully connected layer.	CO2	K1
20.	Define the term Huber Loss.	CO2	K1
Part B			
1.	Discuss in detail about different layers of Convolution Neural Network architecture along with real time example.	CO2	K2
2.	A self-driving car uses a CNN to recognize traffic signs from road images. The CNN applies convolution operations to detect important visual features before classification. Explain how a convolution operation is performed on an input image using a suitable kernel. Show the computation of the resulting feature map and explain how the extracted features help the vehicle identify traffic signs.	CO2	K3
3.	Discuss the different CNN architectures used in deep learning. Compare their structure, working principles, performance characteristics, and computational efficiency with suitable examples.	CO2	K2
4.	A company is developing a CNN-based facial recognition system that processes millions of high-resolution images daily. To reduce computational complexity and memory	CO2	K3

	requirements, the CNN uses sparse interactions and parameter sharing instead of connecting every input pixel to every neuron. Apply the concepts of Sparse Interactions and Parameter Sharing to explain how the CNN efficiently processes facial images. Illustrate how these techniques reduce the number of parameters, computational cost, and memory usage while maintaining effective feature extraction for accurate face recognition.		
5.	Explain in detail about the types of pooling layers and Highlight the Feature map for each type.	CO2	K2
6.	A medical imaging company develops a CNN-based system for tumour detection and image segmentation. The network uses different convolution variants at various stages to improve feature extraction and image reconstruction. Describe how Strided, Tiled, Transposed, and Dilated Convolutions can be applied in the CNN architecture. Explain the purpose of each convolution variant and its contribution to accurate tumour detection and image segmentation.	CO2	K3
7.	Describe in detail about Non-Linearity Functions and highlight the commonly used nonlinear activation functions in CNN.	CO2	K2
8.	A CNN is developed to classify different types of fruits from images. During training, the model computes prediction errors, updates weights, and uses techniques to avoid overfitting. Apply the concepts of loss functions, regularization, optimizers, and gradient computation in the training process of the CNN and explain how each contributes to improving classification accuracy.	CO2	K3

Unit 3: Deep Learning Algorithms For AI

S.No	Questions	CO	BL
Part A			
1.	Define artificial neural network and list its advantages.	CO3	K1
2.	What is Linear Associator model?	CO3	K1
3.	Draw the simple structure of perceptron.	CO3	K1
4.	Define Multilayer Perceptron.	CO3	K1
5.	What is backpropagation and why is it important?	CO3	K1
6.	How backpropagation works?	CO3	K1
7.	Write the Training algorithm of Back propagation Neural Network.	CO3	K1
8.	State any two characteristics of a Hopfield Network.	CO3	K1
9.	List some important points about Boltzmann Machine.	CO3	K1
10.	How do Restricted Boltzmann Machines work?	CO3	K1
11.	What are the types of Restricted Boltzmann Machine	CO3	K1

	(RBM)?		
12.	Define Variational Autoencoders.	CO3	K1
13.	What is epoch and Batch Size?	CO3	K1
14.	Define Associative memory.	CO3	K1
15.	What are Discrete Networks in Hopfield networks and list its categories.	CO3	K1
16.	What is Energy Function in Hopfield Network? State its significance.	CO3	K1
17.	List the steps involved in testing algorithm for Boltzmann Machine.	CO3	K1
18.	What are the Objectives of Boltzmann Machine?	CO3	K1
19.	What is the role of hidden units in a Boltzmann Machine?	CO3	K1
20.	What is the primary purpose of an Autoencoder in deep learning?	CO3	K1
Part B			
1.	Explain in detail about Artificial Neural Network with the neat diagram.	CO3	K2
2.	A library management system uses a neural network to associate book identification numbers with book titles. When a book ID is entered, the system retrieves the corresponding title from memory. Explain how a Linear Associative Network can be used to establish and recall associations between book IDs and book titles.	CO3	K2
3.	A handwritten digit recognition system uses a Backpropagation Neural Network trained on the MNIST dataset. Explain the architecture of the network with a neat diagram and describe how the Backpropagation algorithm trains the network to classify handwritten digits.	CO3	K2
4.	A college develops a system to determine whether students are eligible for a scholarship. The decision is based on attendance, academic performance, and extracurricular achievements. A Perceptron is used to make the classification. Explain the role of a Perceptron in classifying students as eligible or not eligible for a scholarship.	CO3	K2
5.	A pattern recognition system uses a Hopfield Network to store and retrieve binary patterns. Explain the architecture and working of the Hopfield Network with a neat diagram. Discuss how the network stores patterns, retrieves memories, and reaches a stable state.	CO3	K2
6.	An online shopping company uses a Boltzmann Machine to analyze customer purchase histories and identify common buying patterns. Explain how a Boltzmann Machine helps discover hidden patterns in customer purchasing behavior.	CO3	K2
7.	A bank uses a Deep RBM to analyze transaction records and identify unusual activities that may indicate fraud.	CO3	K2

	Explain how a Deep RBM helps detect hidden patterns associated with fraudulent transactions.		
8.	Describe in detail about Variational Autoencoders and Deep Backprop Networks.	CO3	K2

Unit 4: Data Science and Deep Learning

S.No	Questions	CO	BL
Part A			
1.	List the five distinct stages of Data science's lifecycle.	CO4	K1
2.	How performance of deep learning is measured?	CO4	K1
3.	Define Latency and Throughput.	CO4	K1
4.	How Baseline models serve as benchmarks for trained models?	CO4	K1
5.	What are the steps involved in Building a baseline model?	CO4	K1
6.	List the basic approaches to choose hyperparameters.	CO4	K1
7.	Define the term Manual Hyperparameter Tuning.	CO4	K1
8.	Differentiate Grid Search and Random Search.	CO4	K2
9.	What are Debugging strategies?	CO4	K1
10.	List the Important debugging tests.	CO4	K1
11.	List some data science tools.	CO4	K1
12.	What is the goal of manual hyperparameter?	CO4	K1
13.	What is automatic hyperparameter optimization algorithms?	CO4	K1
14.	How to work with data variables and data science tools?	CO4	K1
15.	Define Rule based method.	CO4	K1
16.	What is Data Science?	CO4	K1
17.	Define Modelling.	CO4	K1
18.	How size of the model is calculated in network?	CO4	K1
19.	How overfitting occurs in hyperparameter?	CO4	K1
20.	How to improve the accuracy of the approximation by using the centered difference?	CO4	K1
Part B			
1.	A healthcare organization wants to predict the likelihood of patients developing chronic diseases using historical medical records, laboratory reports, and lifestyle information. A Data Scientist is assigned to manage the project from data collection to model deployment. Apply the fundamentals of Data Science to explain how the healthcare organization can transform raw patient data into meaningful insights. Discuss the responsibilities of the Data Scientist at each stage of the data science lifecycle, including data collection, preprocessing, analysis, model building, evaluation, and decision-making.	CO4	K3
2.	Illustrate the lifecycle of a data science with neat diagram.	CO4	K2

3.	Explain data science tools for working with diverse data variables.	CO4	K2
4.	A university maintains a database containing student information such as roll number, department, CGPA, attendance percentage, and placement status. A data scientist uses various tools to analyze the data and identify factors affecting student placements. Explain the role of data variables and data science tools in organizing and analyzing student data.	CO4	K3
5.	Describe data modelling and featurization in ML.	CO4	K2
6.	A smart city project gathers data from traffic cameras, sensors, and weather stations to improve traffic management. Apply deep learning techniques to explain how the collected data can be processed and analyzed as part of a data science workflow to support intelligent traffic management and decision-making.	CO4	K3
7.	A hospital collects patient data, including age, blood pressure, cholesterol levels, and disease status. Apply appropriate data visualization techniques to present the patient data and explain how the visualizations help healthcare professionals identify trends and make informed decisions.	CO4	K3
8.	Explain how data visualization helps data scientists.	CO4	K2

Unit 5: Applications of Deep Learning

S.No	Questions	CO	BL
Part A			
1.	State the importance of Chest X-ray Image Analysis.	CO5	K1
2.	List the steps involved in the detection of chest x-ray image.	CO5	K1
3.	What is Object detection and classification?	CO5	K1
4.	List the three modules in R-CNN.	CO5	K1
5.	List the importance of RGB and Depth Fusion.	CO5	K1
6.	What are the approaches in Natural Language Processing?	CO5	K1
7.	List the applications of NLP.	CO5	K1
8.	What are the Significances of Dimensionality Estimation in Data Analysis?	CO5	K1
9.	State the term Principal Component Analysis (PCA).	CO5	K1
10.	Name the metrics for measuring forecasting accuracy.	CO5	K1
11.	What is Deep Reinforcement Learning?	CO5	K1
12.	Define Tokenization.	CO5	K1
13.	What is the role of robots in manufacturing and logistics?	CO6	K1
14.	How does deep learning support decision-making in real-world applications?	CO6	K1

15.	What is the role of deep learning in maximizing charity donations?	CO6	K1
16.	List Text Generation applications.	CO6	K1
17.	What is Machine Translation?	CO6	K1
18.	What is Time Series Forecasting?	CO6	K1
19.	List the Challenges in Traditional Power Grids.	CO6	K1
20.	What are the benefits of Smart Grids?	CO6	K1
Part B			
1.	An online shopping platform allows users to upload images of products. The system detects the product in the image and classifies it into the appropriate category for product search. Explain how Object Detection and Classification can be used to identify products from uploaded images and improve the customer shopping experience.	CO5	K3
2.	Explain the concept of RGB and depth image fusion with a neat block diagram. Describe the process of combining RGB and depth information, discuss its advantages, limitations, and explain its applications in computer vision and autonomous systems.	CO5	K2
3.	Explain the major Natural Language Processing (NLP) tasks used in deep learning. Discuss the objectives, working principles, advantages, and practical applications of text classification, sentiment analysis, machine translation, and named entity recognition.	CO5	K2
4.	An electricity distribution company wants to predict future power demand using historical electricity consumption data. Apply deep learning-based time series forecasting techniques to explain how accurate demand prediction helps in efficient energy management and resource allocation. Discuss the forecasting process, suitable deep learning models, and benefits.	CO5	K3
5.	A meteorological department collects daily temperature, rainfall, and humidity data over several years. The department wants to predict future weather conditions. Explain time series forecasting methods to predict future weather patterns using historical climate data. Explain how the forecast can assist in disaster preparedness and agricultural planning.	CO6	K2
6.	Explain in detail about Dimensionality Estimation in Data Analysis.	CO6	K2
7.	A city is developing a smart electric power grid that integrates solar panels, wind turbines, battery storage systems, and conventional power plants. The grid must continuously monitor energy demand and automatically control the distribution of power from different energy	CO6	K3

	sources to ensure reliability and efficiency. Apply suitable data science and machine learning techniques to explain how controllable energy resources can be managed in the smart power grid. Discuss how data analysis and predictive models can improve energy distribution and grid stability.		
8.	A charity platform and an industrial automation company are implementing Deep Neural Networks to improve their services. Describe how Deep Neural Networks can be applied to maximize charity donations and enhance robotic control in industrial environments. Explain the advantages of deep learning in both applications.	CO6	K3

Course Coordinator

HOD